



CHANHENG MENH

ABOUT ME

I'm a Digital Infrastructure graduate from the American University of Phnom Penh with hands-on experience in AI/ML, DevOps, and IT operations. I build and deploy machine learning models, provision cloud infrastructure with Terraform, and maintain CI/CD pipelines — end to end, without needing a team to glue the pieces together. I'm seeking roles in data, AI/ML, DevOps, or IT where technical ownership and problem-solving matter.

CONTACT

- +85570494393
- chanhengmenh@gmail.com
- github
- linkedin

EDUCATION

2023-2026
American University of Phnom Penh
Bachelor of Science in Digital Infrastructure

LANGUAGES

- Khmer:** Native
- English:** Professional Working Proficiency
- Chinese:** Basic (Elementary)

SKILLS

Infrastructure & DevOps

- Terraform
- Jenkins
- Docker
- Kubernetes
- AWS (EC2, S3, RDS)
- GitHub Actions
- SonarQube
- Grafana
- Prometheus
- Linux

Data & AI/ML

- Python
- PyTorch
- TensorFlow
- Scikit-learn
- YOLO
- Optuna
- Pandas
- NumPy
- Computer Vision
- NLP

Databases & Cloud

PostgreSQL, MongoDB, DynamoDB,
Firebase, Supabase, SQL

Development & APIs

FastAPI, SQLAlchemy, Node.js,
RESTful API Design, Git

REFERENCES

Dr. Tek Ming Ng

Dean, Faculty of Digital Technologies, AUPP
n.ming@aupp.edu.kh, [linkedin](#)

Dr. Kuntha PIN

Lecturer, Faculty of Digital Technologies, AUPP
k.pin@aupp.edu.kh, [linkedin](#)

TECHNICAL PROJECTS

NLP Discord Moderation System | RoBERTa, PyTorch, Optuna | [github](#)

- Engineered a deep learning natural language processing pipeline utilizing the RoBERTa architecture for automated Discord text moderation and sequence classification.
- Optimized model performance by implementing Optuna for automated hyperparameter tuning (learning rate, batch sizes, epochs), successfully deploying the optimized classifier for targeted text analysis.

Public School Management System | FastAPI, Next.js, PostgreSQL, Vercel, Railway | [github](#)

- Architected a high-availability educational platform with a Next.js frontend and a FastAPI backend, ensuring high service reliability.
- Engineered a custom behavior analytics engine to track real-time engagement and a secure REST API featuring JWT authentication and RBAC.
- Optimized data infrastructure by configuring internal PostgreSQL connections on Railway, eliminating public network latency.

Spam Classification Pipeline | Scikit-learn, Pandas, Python | [github](#)

- Architected a traditional machine learning pipeline evaluating both tree-based (Random Forest, Gradient Boosting) and linear (Logistic Regression, Naive Bayes) classification models.
- Implemented distinct preprocessing workflows for tree-based vs. linear architectures, optimizing model selection based on F1-score performance for imbalanced datasets.

AUPP Intelligent Chatbot | Llama 3.1, Ollama, ChromaDB, Streamlit, Unsloth | [github](#)

- Built a fully local, cost-free university chatbot for AUPP by combining RAG, knowledge distillation, and fine-tuning — eliminating dependency on external APIs while maintaining high response accuracy grounded in real website content.
- Fine-tuned Meta-Llama-3.1-8B-Instruct on 2,500+ LLM-distilled Q&A pairs using Unsloth + LoRA (4-bit quantization), then deployed the GGUF-exported model locally via Ollama with a Streamlit UI featuring streaming responses and source citations.

EXPERIENCE

SpaceGap

May 2025 - Jul 2025

UI/UX & Frontend Developer Intern

- Collaborated in a team of 5 to design and deploy 2 web application pages, contributing to UI/UX design in Figma for a student management platform.
- Designed key platform modules including student dashboard, event scheduling, reports and analytics, user profile, and notification systems.
- Developed interactive frontend components, calendar, and to-do list using Bootstrap for responsive UI architecture.

Spring Education Centre

Jul 2023 - Aug 2023

Digital Content Creator Intern

- Produced and published weekly video content across TikTok and Facebook to drive brand awareness for an early-stage education startup.
- Reached over 4,000 viewers per video, supporting the centre's initial community growth and audience engagement.

LEADERSHIP & INVOLVEMENT

Chess Club President, AUPP | Jan 2026 – May 2026

- Facilitated weekly sessions for 15 to 20 members, cultivating a campus culture of analytical problem-solving, and strategic thinking.
- Directed the planning and execution of the AUPP Chess Tournament, managing event logistics, player brackets, and promotional outreach.

Mentor - AUPP Student Affairs Mentorship Program | Sep 2025 - May 2026

- Guided incoming freshmen through their university transition and academic planning, while delivering targeted academic support in AI and machine learning coursework.

Member - Global Shapers Phnom Penh Community | Jan 2024 - Present

- Led end-to-end digital campaigns for the Riel Hackathon for Kids, scaling event visibility and impact.
- Coordinated site logistics, vendor management, and cross-functional outreach for community events.

IT Support Volunteer - 20th CamTESOL | 2024

- Managed Audio/Visual (A/V) setup and provided live technical troubleshooting for a dedicated presentation room during a major international conference.